VINCENT XAYASAK

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ML and software engineering graduate with LLM, Agentic AI, and model evaluation research experience.

EDUCATION

University of California, Davis | Davis, CA *June 2026*
B.S. in Computer Science | GPA: 3.5

Relevant Coursework: Machine Learning (supervised/unsupervised learning, SVM, kernels, PCA, bias-variance trade-offs), Software Engineering (OOP design patterns, CI/CD, unit testing, containerized architectures), Computer Security & Privacy (encryption, authentication, access control, network security)

PROFESSIONAL EXPERIENCE

Advantest | San Jose, CA *June 2025 – September 2025*
AI/ML Intern

- **Fine-tuned Llama 3.3 70B via QLoRA** for SmarTest8 syntax; scaled from 4GB to 48GB VRAM after delivering a high-impact proof-of-concept.
- **Engineered a RAG pipeline** with `BAAI/bge-m3` embeddings to eliminate hallucinations and accelerate hardware specification retrieval.
- **Deployed a local FastAPI web service** with a custom request queue to manage concurrent inference and optimize model throughput.
- **Automated firmware integration** via LLM scripts, reducing engineering turnaround from one week to 5 hours (∼90% efficiency gain).
- **Synthesized technical documentation** for semiconductor validation standards, ensuring AI-generated automation adhered to V93000/SmarTest8 architecture.

RESEARCH

BRIDGE: Bias Register-Integrated Dataset for LLM Bias Evaluation | Davis, CA *January 2026 – May 2026*
Research Assistant

- **Built a 19,421-sentence benchmark** spanning formal institutional and informal social text under a unified harmful/harmless/antibias taxonomy from seven open-source corpora spanning 150 years of discourse.
- **Curated formal-text datasets** from `Jim Crow Laws`, `Pile of Laws`, and `GovReport`; engineered a Gemini API pipeline for policy categorization with expert-adjudicated manual verification.
- **Designed GRDC**, a five-metric diagnostic suite evaluating register-aware bias detection across eight frontier LLMs, identifying the Register Polarity Flip failure mode (DC=1.0, $p \approx 6 \times 10^{-5}$).
- **Executed few-shot/zero-shot evaluations** measuring racial, gender, and political bias dimensions; demonstrated RPF replicates on independent external datasets and is not mitigated by few-shot prompting.
- **Co-authored paper in submission** to NeurIPS 2026 Evaluations & Datasets Track on register-stratified bias evaluation for institutional LLM auditing workflows.

Measuring the Shape of Reasoning-Model Behavior | Davis, CA *January 2026 – May 2026*
Research Assistant

- **Introduced Reasoning Path Consistency (RPC)**, a three-axis metric (lexical, structural, semantic) quantifying similarity of reasoning traces across repeated attempts, validated against human judgments (Spearman ρ=0.34).
- **Developed Parametric Sampling Fingerprints (PSF)** fitting Pass@k trajectories with three-parameter curves (mean R^2=0.96), summarizing how model accuracy scales with additional sampling attempts.
- **Evaluated 16 reasoning models** across four benchmarks (law, mathematics, graduate science, multi-modal); Pass@1-matched models diverge up to 0.17 in semantic RPC and 10 points on Pass@4.
- **Demonstrated RPC and PSF are statistically independent**, capturing distinct dimensions of model behavior that scalar Pass@1 leaderboard accuracy conceals.
- **Co-authored paper in submission** to ACL on multi-dimensional LLM capability measurement beyond single-number benchmark accuracy.

SKILLS & INTERESTS

Languages: Python, Java, C/C++, Go, SQL, JavaScript, HTML/CSS
AI/ML Architecture: PyTorch, Hugging Face, LLM Fine-tuning (LoRA/QLoRA), RAG, PEFT, ChromaDB
AI Skills: RL (GRPO/PPO), On-Policy Distillation, Adversarial Benchmarking, Bias Evaluation, Model Reasoning Evaluation
Dev & MLOps: Docker, Kubernetes, FastAPI, AWS, Git, BitBucket, Cursor, Claude Code, MCP
Inference & Optimization: CUDA, Quantization (GGUF/AWQ), Concurrent Inference Pipelines
Interests: LLM Evaluation, Model Reasoning & Bias, MLOps, NLP, Software Architecture